

Flash with GP2040-CE (specific version for WAVESHARE PICO ZERO - normal PICO version will work but will auto assign various GPIO etc just confusing things)

Load webconfig by holding b3+b4+s2 (game controllers settings in windows will show you what buttons are 3, 4 and 8 as you press them, it's a good time to get used to using this - after you have configured it b3+b4+s2 will be different). Webconfig then serves from the pico on 192.168.7.1

EVERY TIME YOU MAKE A CHANGE IN A SECTION CLICK SAVE, AND GO TO REBOOT - UNTIL YOU ARE FINISHED SELECT 'WEB CONFIG' AS YOUR REBOOT OPTION SO THAT YOU CAN GO TO 197.168.7.1 AGAIN AND CONTINUE SETUP!

Configuration > GPIO Pin Mapping

Profile 1 - GPIO Pin Mapping

Profile name

Profile 1

Max 16 characters. Letters, numbers, and spaces allowed.

GP0	S1 x	x	GP15	Select...
GP1	S2 x	x	GP16	Select...
GP2	R1 x	x	GP17	Select...
GP3	B4 x	x	GP18	Select...
GP4	B2 x	x	GP19	Select...
GP5	Select...		GP20	Select...
GP6	Select...		GP21	Select...
GP7	Select...		GP22	Select...
GP8	L2 x	x	GP23	Select...
GP9	L1 x	x	GP24	Select...
GP10	B3 x	x	GP25	Select...
GP11	B1 x	x	GP26	Assigned to ac
GP12	Select...		GP27	Select...

Set as shown, clear all others I've assigned them left to right b1, b2, b3, b4, l1, r1, l2 - and s1 and s2 are the tactile swiches

Gamepad USB Host Addon

USB host not enabled!

Rotary Encoders

Encoder 1

Enabled

Pin A/CLK

28

Pin B/DT

27

Pulses/Revolution (PPR)

In configuration -> Add-ons Configuration

Rotary Encoders

Pin A and Pin B are GPIO 28 & 27, depending on which side of the turntable you place your hand up and down may be inverted for you, in which case switch these numbers over

Multiplier & Reset After {ms} Multiplier is down to personal preference, but likely doesn't need to be changed (I've only used sim files in Project Outfox. Reset After sets the up/down of the turn table back to neutral after using it, otherwise if it is disabled (set to 0) it would continue to register the last direction input until engaged again, which likely isn't hat you want.

Control Mode - the home controller, arcade controller, even DJ hero, use an optical sensor of sorts and merely acts as an up and down switch, so D-Pad up and down makes the most sense to me, but, you can set it to analogue if for some reason incremental input would be useful.

Allow Wrap-Around isn't needed here, and if I click this the whole page goes goofy and won't allow saving or anything. Just avoid it for now.

Reactive LEDs

LED #0 LED GPIO Pin 5	LED #1 LED GPIO Pin 6	LED #2 LED GPIO Pin 7	LED #3 LED GPIO Pin 12	LED #4 LED GPIO Pin 13
Action B2	Action B4	Action R1	Action L2	Action L1
Default Mode Static On	Default Mode Static On	Default Mode Static On	Default Mode Static On	Default Mode Static On
Active Mode Static Off	Active Mode Static Off	Active Mode Static Off	Active Mode Static Off	Active Mode Static Off
LED #5 LED GPIO Pin 14	LED #6 LED GPIO Pin 15	LED #7 LED GPIO Pin 5	LED #8 LED GPIO Pin -1	LED #9 LED GPIO Pin -1
Action B3	Action B1	Action S1	Action None	Action None
Default Mode Static On	Default Mode Static On	Default Mode Static Off	Default Mode Static Off	Default Mode Static Off
Active Mode Static Off	Active Mode Static Off	Active Mode Static On	Active Mode Static On	Active Mode Static On

Save LEDs

On the same page is the **reactive LEDs** section - set as shown for GPIO and Action, as those are the GPIO's I chose making the PCB. In the picture it shows I have set the LEDs to be on until 'active' (pressed) at which point they will go off. Usually we do the opposite of this, you can if you want, just get the GPIO and 'Action' (assigned button) right as it is here.

RGB LED Configuration

Data GPIO Pin (-1 for disabled) 26	LED Format GRB	LED Layout Custom
LEDs Per Button 1	Max Brightness 255	Brightness Steps 10

☐ Turn Off When Suspended

RGB LED Button Order

Configuration > LED Configuration

This section can be a bit hard to understand, just duplicate what is shown here - make sure there are no buttons listed under ‘Assigned Buttons’

A1

A2

Player LEDs

Player LED Type

Off

▼

Turn Off When Suspended

Case RGB LEDs

Color Type

Ambient

▼

RGB LED Index

0

▼

RGB LED Count

2

▼

For Case RGB LEDs, set a starting index and the case RGB count. The index must be after the last LED button defined by

Save

Lower on the same LED page. I haven’t documented how to make animations etc here but you can modify these and switch with them with hotkey combos documented in the GP2040-CE docs. For now just having it working will do, so copy this here,.

On this last reboot you can select to reboot as a controller. Badabing me ol’ china